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A Hybrid Machine Learning Approach with Explainable AI for Prostate Cancer Diagnosis

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***Abstract*—Prostate cancer is the most common nonskin cancer in the human body. Early-stage prostate cancer frequently has no symptoms and progresses slowly, necessitating only vigilant surveillance. One of the most prevalent malignancies in males is prostate cancer, and successful treatment planning depends on early detection. The stacking-based ensemble learning framework proposed in this article combines several fine-tuned machine learning models, such as Random Forest (RF), XGBoost (XGB), Gradient Boosting (GB), and LightGBM (LGBM), SVM, Logistic Regression (LR) etc. The dataset was normalized using MinMax scaling and outlier elimination using the interquartile range approach. GridSearchCV with stratified cross-validation was used to optimize the hyperparameters. The system leveraged complementary capabilities of individual classifiers by combining the optimized base models in a stacking ensemble with RF, XGB, GB, LGBM with soft voting. In comparison to single models, ensemble models give experimental evaluation on the prostate cancer dataset, achieving accuracies in range of 85% to 96%. Key predictive biomarkers were found by interpreting the models using XAI approaches (SHAP and LIME). According to the findings, the suggested framework is a potential decision-support tool for prostate cancer risk prediction since it combines PCA, ensemble learning, and XAI to give both strong accuracy and transparency.**

***Index Terms*—Prostate Cancer, PCa, Stacking, Ensemble, XAI, Cancer, Soft Voting, Machine Learning**

I. Introduction

Prostate cancer is the fifth most prevalent cause of cancer-related death for males worldwide, making it a serious global health concern. Due to economic expansion and an aging global population, this disease, which has a significant impact, is expected to continue to rise [1]. Mortality rates vary significantly between nations, while it is the most common cancer in men diagnosed worldwide, after lung cancer [2]. Between 1975 and 1991, the prostate cancer death rate increased before starting to decline. Between 1991 and 2008, the death rate decreased by 39 percent. The annual death rates from 2006 to 2008 are somewhat lower than those from 1975. It is unknown what caused the increase in mortality between 1975 and 1991. [3]. The incidence of detectable metastases at diagnosis has decreased from over 50% of cases in the 1970s to less than 10% today, but the prevalence of PCa has increased and the stage at which it is usually diagnosed has changed in western countries due to routine PSA testing. [4]. Prostate cancer has the highest incidence and mortality rates in Africa, where this prevalence is most noticeable [5].Pro-inflammatory cytokines, myeloid cells that infiltrate the tumor, and immune cells are significant players in almost every stage of the development of inflammation-induced cancer. An increasing number of studies have established a causal link between inflammation and tumors. In addition to supporting the proliferation and survival of cancer cells, the inflammatory response of the tumor microenvironment is essential for tumor angiogenesis, metastasis, immune evasion, and treatment resistance. [6]. In 2018, 1,276,106 new cases were reported worldwide, with wealthier countries showing a greater incidence [7]. Even though prostate cancer is the third most frequent disease diagnosed worldwide, after lung and breast cancer, it is anticipated that 33,330 people will die from it in the United States alone in 2020 [8]. Little understanding of PCa insights is required to build a machine learning model; organized datasets including input variables and outputs are sufficient. For example, a number of new technologies, including genomics, magnetic resonance imaging (MRI), and biomarkers (exosomes and molecular imaging), are available for the screening and diagnosis of PCA [9]. In order to decrease unnecessary biopsies and diagnose indolent malignancies, prostate cancer risk calculators have been developed based on the addition of basic clinical characteristics to PSA. The more widely used European Randomized Study of Screening for Prostate Cancer (ERSPC) risk calculator (RC) and any ultrasonic abnormalities in the prostate to improve the prediction [10]. In order to predict prostate cancer, this study suggests a stacking ensemble that combines Random Forest, XGBoost, Gradient Boosting, and LightGBM. To improve accuracy, the improved models

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were incorporated following preprocessing and hyperparameter adjustment. The model's potential for early prostate cancer diagnosis is shown by the experimental results, which show great predictive performance. The main contributions of this paper are as follows:

- Creation of a reliable stacking classifier for prostate cancer prediction that ensemble Random Forest, XGBoost, Gradient Boosting, and LightGBM.
- To enhance model performance, thorough preprocessing is used, which includes controlling class imbalance, Min-Max scaling, and outlier elimination.
- Using grid search and cross-validation, each base model's hyperparameters are extensively adjusted to maximize prediction accuracy.
- Establishing a final ensemble model that improves classification performance by utilizing each base classifier's advantages.
- High prediction accuracy and efficiency are demonstrated on datasets related to prostate cancer, offering a useful framework to help doctors diagnose the disease early.

The work concludes with a demonstration of the approach on three distinct real-world datasets, highlighting its versatility and effectiveness across multiple domains.

II. Related Work

Due to stringent privacy and security regulations, as well as the intricacy of data collecting and processing, clinical data are frequently restricted. As a result, only publicly accessible datasets or those acquired through research data agreements were utilized. Health records, MRI and ultrasound pictures, and numerical measures of afflicted areas are only a few examples of the various types of medical data that contribute to their overall. In addition to determining the disease's risk, researchers are constantly building diagnostic models for PCa that help doctors make the best treatment choices by forecasting the disease's course. Laabidi et al. [11] predicted outcomes on two datasets: Using both standardized and non-normalized data, models such as RNN, LR, Naive Bayes, SVM, RF, DT, KNN, and XGBoost were assessed on diabetic and prostate cancer datasets. Accuracy, AUC, and F1-score were used to gauge performance. On the PCa dataset, the RNN model with three hidden layers and tanh activation produced the highest results: 81% accuracy, 0.866 AUC, and 0.809 F1-score. Akinnuwesi et al. [12] Due to inconsistent recording and early symptoms that overlap with other illnesses, PCa data are frequently inaccurate. To address this, SelectKBest, chi-squared, and PCA were applied for feature reduction after balancing the dataset through up- and down-sampling. The SVM-PCa-EDD model achieved an average classification accuracy of 90%. Wang et al. [13] proposed a diagnostic technique based on ultrasound video data gathered from 501 patients—276 of whom had PCa and 225 of whom had benign lesions. A machine-learning model was trained using a random sample of 231 individuals, 118 of whom had cancer and 113 of whom were benign. 851 characteristics were taken from transrectal ultrasound footage. Only the 14 most significant features were chosen after the dimensionality of the data was reduced using the LASSO approach. Based on these chosen features, a PCa prediction model was constructed utilizing SVM and RF algorithms. The SVM and RF models' diagnostic capabilities were contrasted with the MRI-based diagnoses made by specialized doctors. To assess the effectiveness of diagnostic support, a novel approach called MLAD was put out that combines the findings from physician diagnoses with machine learning models. The range of the AUC findings was 0.65 to 0.81.

Qian et al. [14] presents a unique approach for PCa detection based on MRI imaging termed ProCDet, which also uses MRI image data. Prostate classification using a concentration mechanism-centered classification algorithm, PCa disease classification using a 3D segmentation network based on Focal Tversky Loss, and MRI image registration are the three primary modules of this approach.Utilizing unlabeled data, a self-supervised network helps identify key characteristics in the pictures. The algorithms are trained and tested using MRI data from several patients. According to the results, ProCDet has an average of 0.6275 false positive lesions per patient and a true positive rate of up to 91.82%. This approach's scientific accomplishments include lowering physician burden, increasing diagnostic accuracy, and being the first to employ self-supervised learning to improve cancer detection efficacy.

Wang et al. [15] PCa study made use of patient data from the MIMIC-IV database. First, the data was cleaned, feature selection, outlier processing, and missing data handling. LR, SVM, Deep Neural Network (DNN), XGBoost, LGBM, and CatBoost were among the machine learning techniques used to construct predictive models. These models were assessed using ROC curve analysis and cross-validation. In a 4:1 ratio, the data were divided into training and test sets. Features that have missing values greater than forty percent were disqualified. Recursive Feature Elimination (RFE) was utilized for feature selection, while the Isolation Forest technique was utilized to identify outliers. The model was trained using five-fold cross-validation. According to evaluation results based on AUC value 0.93, LGBM performed the best.

III. Methodology

This section presents the methodological framework followed in this study. The approach involves data preprocessing, model development with hyperparameter optimization, and the construction of an ensemble classifier using stacking. The overall workflow is designed to ensure robust and accurate classification of prostate cancer cases in figure 1.

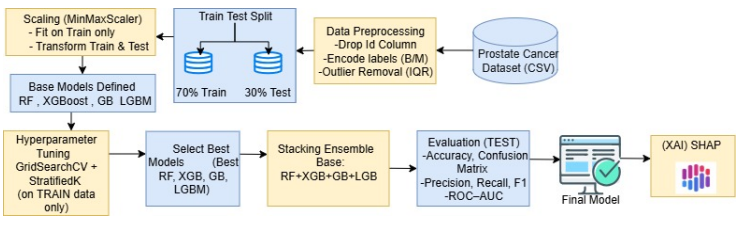


Fig. 1. Proposed Methodology Technique

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A. Dataset

The study's dataset, which includes clinical parameters for the detection of prostate cancer, was taken from the Kaggle open-source repository [16]. It includes a binary diagnostic result and 100 samples with 10 attributes. With 62 benign and 38 malignant instances, the target variable *diagnosis_result* was encoded into binary classes: Benign (B) mapped to 0 and Malignant (M) mapped to 1. Traditional machine learning techniques were used to guarantee strong and trustworthy predictions because deep learning approaches run the danger of overfitting due to the dataset's short size. The correlation between the features is shown in Figure 2.

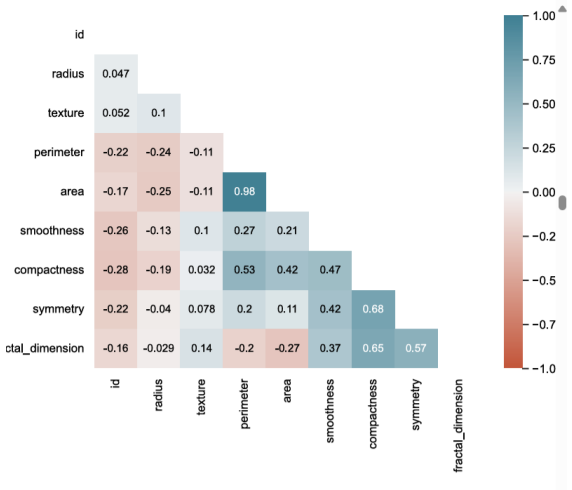


Fig. 2. Correlation Heatmap.

B. Data Preprocessing

To guarantee the consistency and dependability of the input features, the dataset was initially pre-processed.The following steps were applied:

1) **Column Removal:**The *id* attribute was eliminated since it has no bearing on the diagnosis.
2) **Target Encoding:**Benign (B) $\rightarrow$ 0 and malignant (M) $\rightarrow$ 1 were the binary values that were created from the categorical target variable *diagnosis_result*.
3) **Outlier Detection:** To prevent distortion in training, outliers were identified and eliminated using the Interquartile Range (IQR) technique.
4) **Feature Normalization:** All numerical features were converted into the range $[0, 1]$ using Min-Max Scaling, guaranteeing equal weighting throughout model training. The formula is given below in equation 1:

$$X_{scaled} = \frac{X - \min(X)}{\max(X) - \min(X)} \tag{1}$$

C. Machine Learning Techniques

Four machine learning classifiers were trained as base learners for the ensemble model with soft voting and stacking classifiers. The equation 2, 3 are given below:

- **Random Forest (RF):** The final output is generated by aggregating predictions from a collection of decision trees called Random Forest. Majority voting is used to classify and the formula is given below:
  - **Classification (majority voting):**

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_T(x) \tag{2}$$

  - **Regression (averaging):**

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} h_t(x)\} \tag{3}$$

- **XGBoost:** By merging several decision trees and reducing a regularized loss function, XGBoost makes predictions, it reduce the chance of overfitting and the formula is given below in equation 4 :

$$\text{Loss} = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{4}$$

where l is the loss function, $\hat{y}_i$ is the predicted output, f_k represents the k-th tree, and $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ is a regularization term controlling tree complexity.
- **Gradient Boosting (GB):** A boosting method that builds sequential trees, each correcting errors of the previous one. By gradually training weak learners usually decision trees to minimize a differentiable loss function, gradient boosting creates an additive model. At iteration m, the forecast is given below in equation 5:

$$\hat{y}^{(m)} = \hat{y}^{(m-1)} + \nu \cdot h_m(x) \tag{5}$$

where $\hat{y}^{(m-1)}$ is the previous iteration's prediction, $h_m(x)$ is the m-th weak learner fitted to the negative gradient of the loss function, and ν is the learning rate controlling the contribution of each tree. The objective is to reduce the overall loss the equation is give in equation 6.

$$\text{Loss} = \sum_{i=1}^{N} l(y_i, \hat{y}_i) \tag{6}$$

where $l(y_i, \hat{y}_i)$ is the chosen loss function for the training data.
- **LightGBM (LGBM):** A high-performance gradient boosting framework that grows trees leaf-wise for efficiency and accuracy. The gradient boosting technique LGBM minimizes a regularized loss function by growing trees leaf-by-leaf and the formula is given below in equation 7:

$$\text{Loss} = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{7}$$

with $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T} w_j^2$, where T is the number of leaves in tree f_k, and w_j is the leaf weight. This strategy improves accuracy and efficiency for high volume datasets.

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D. Hyperparameter Tuning

Each base model underwent hyperparameter optimization using **Grid Search with Stratified 5-Fold Cross Validation**. The parameter search space included:

- **Random Forest:** Number of trees, maximum depth, minimum samples per split.
- **XGBoost:** Number of estimators, learning rate, maximum depth.
- **Gradient Boosting:** Number of estimators, learning rate.
- **LightGBM:** Number of estimators, learning rate.

The best hyperparameters were selected based on cross-validation accuracy.The parameter used in these algorithms have been given in table I.

TABLE I
OPTIMIZED HYPERPARAMETERS OF BASE MODELS AND ENSEMBLE

Classifier	Optimized Parameters
Random Forest (RF)	n_estimators=100, max_depth=None
XGBoost (XGB)	n_estimators=100, learning_rate=0.1, max_depth=3
Gradient Boosting (GB)	n_estimators=100, learning_rate=0.1
LightGBM (LGBM)	n_estimators=100, learning_rate=0.05
Stacking Ensemble	Meta-learner=RF, CV folds=5
Soft Voting	Voting=Soft, Weights=[1,1,1,1]

E. Stacking Ensemble Construction

The stacking ensemble combined the tuned based learners:

1) Predictions from RF, XGBoost, GB, and LGBM were generated as meta-features.
2) A **Random Forest classifier** was used as the meta-learner to integrate these predictions and produce the final output.
3) Stratified 5-Fold cross validation was applied to ensure fair training of the meta-learner and reduce bias.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup

All experiments were conducted on a Windows 11 Home machine with core i3 processor and 32 GB RAM. The implementation was performed in Python using the Scikit-learn, XGBoost, and LightGBM libraries in table II.

TABLE II
SYSTEM CONFIGURATION FOR EXPERIMENTAL EVALUATION

Specification	Details
Processor	Intel® Core™ i3-13100F Processor
RAM	32 GB
Operating System	Windows 11
Implementation	Python (Scikit-learn)

B. Performance Evaluation

The performance of the final stacking ensemble was assessed using the following metrics in figure 3:

- **Accuracy:** Overall proportion of correct predictions the formula is given in equation 8.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{8}$$

- **Precision:** Fraction of correctly identified malignant cases among all predicted malignant cases. The equation is given in equation 9.

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

- **Recall (Sensitivity):** Fraction of correctly identified malignant cases among all actual malignant cases. The equation is given in equation 10

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

- **F1-Score:** Harmonic mean of Precision and Recall, balancing both metrics. The formula is given in equation 11.

$$F1 - score = 2 \times \frac{precision \times recall}{precision + recall} \tag{11}$$

This multi-metric evaluation guarantees a thorough appraisal of the suggested approach's robustness and efficacy.

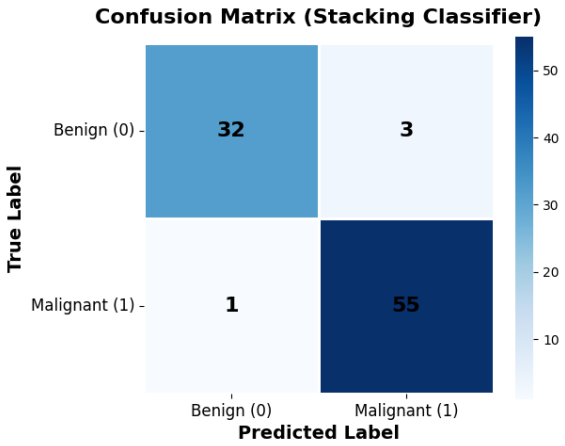


Fig. 3. Confusion Matrix of the Ensemble Model.

C. Experimental Results

Table III shows the performance of individual models, while Table IV summarizes the performance of ensemble models using both hard and soft voting. The ROC curve and Comparison Graph are shown below:

D. Visual Analysis of Results

Classifier performance can also be further understood by visual comparisons utilizing bar charts and ROC curves in addition to numerical performance indicators. Results from Explainable AI (XAI) employing SHAP and LIME are also presented to emphasize interpretability at the feature level.

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TABLE III
PERFORMANCE OF INDIVIDUAL CLASSIFIERS

Classifier	Accuracy	Precision	Recall
CatBoost	0.92	0.92	0.92
Ridge Classifier	0.88	1.00	0.75
Random Forest	0.8529	0.81	1.00
SVM	0.8824	0.84	1.00
Logistic Regression	0.8529	0.81	1.00
LightGBM	0.8235	0.80	0.95
XGBoost	0.8235	0.80	0.95
Extra Trees	0.84	0.88	0.81

TABLE IV
ENSEMBLE MODEL PERFORMANCE

Ensemble Model	Accuracy	Precision	Recall
Stacking (XGB, GB, LGBM, RF)	0.9560	0.96	0.98
Soft Voting (RF, XGB, LGBM)	0.9600	0.93	1.00
Stacking (RF, XGB, SVM)	0.9032	0.86	1.00
Soft Voting (XGB, RF, LGBM, LR, SVM)	0.8485	0.89	0.85
Soft Voting (XGB, RF, LGBM, LR)	0.9200	0.92	0.96
Stacking (XGB, GB, LGBM)	0.9355	0.90	1.00
Stacking (RF, XGB, GB, LGBM)	0.9560	0.95	0.98

1) ROC Curves and Comparison Graphs: Both individual classifiers and ensemble models' ROC curves are displayed. By obtaining higher AUC scores, ensemble models regularly outperform single classifiers, as shown in Figures 4 and 5. Bar comparison plots show the relative accuracy, precision, and recall values for each model Figures 6 and 7.

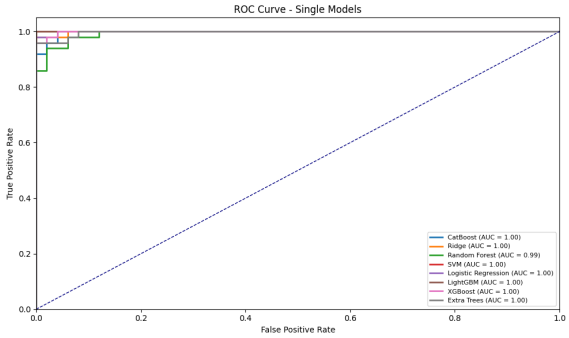


Fig. 4. ROC curves for individual classifiers.

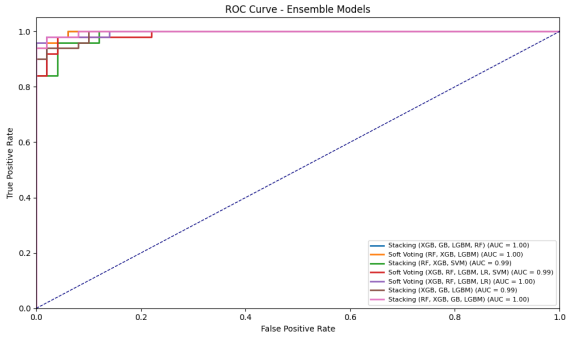


Fig. 5. ROC curves for ensemble models.

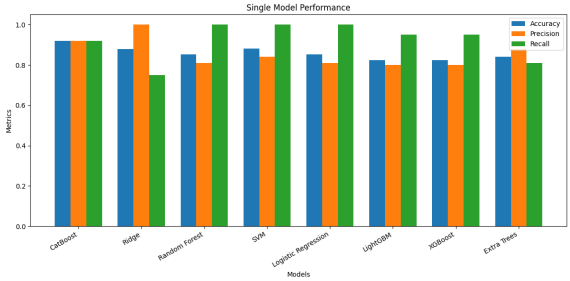


Fig. 6. Performance comparison of individual classifiers.

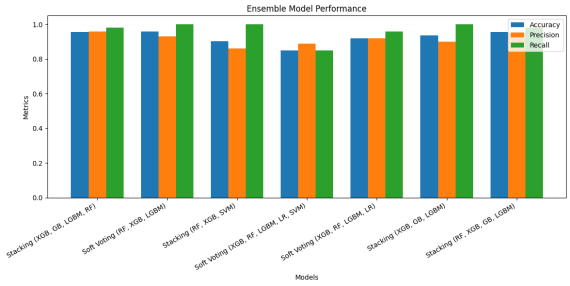


Fig. 7. Performance comparison of ensemble models.

2) Explainable AI (XAI) Visualization: Model predictions were explained using SHAP to increase interpretability. The most significant clinical and demographic characteristics influencing the categorization results are displayed in the SHAP summary plot Figure 8. These revelations improve transparency and give physicians a clearer picture of how the ensemble model makes its predictions.

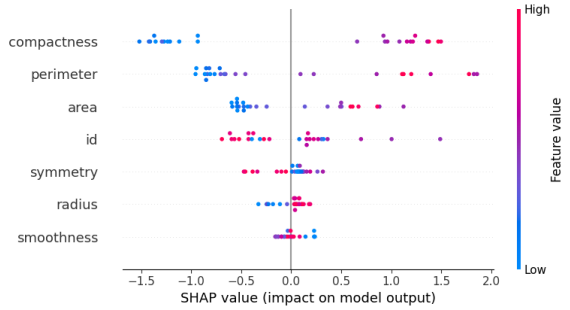


Fig. 8. SHAP summary plot showing feature contributions.

E. Analysis of Results

The various models demonstrated robust classification for PCa detection comparison among different methods and proposed method is presented in Table V, achieving high accuracy (up to 93.94%) with great Precision and Recall values. Soft voting ensembles achieved up to 96% accuracy, regularly outperforming hard voting ensembles. The accuracy gain from ensemble models over individual models demonstrates how well combining several models can reduce individual biases and shortcomings. The viability of SVEM for predictive PCa risk assessment was further supported by the fact that the ensemble technique did not appreciably lengthen training time. The findings show that SVEM offers a dependable and effective predictive model that is appropriate for practical

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clinical settings where precise prostate cancer identification is essential.

TABLE V
PROPOSED MODEL COMPARISON WITH RELATED WORKS

Paper	Modality	Inst.	Method	Acc.
Laabidi et al. [11]	Clinical	–	RNN, ML models	81%
Akinnuwesi et al. [12]	Clinical	–	SVM-PCa-EDD	90%
Wang et al. [13]	Ultrasound	501	SVM, RF	AUC 0.81
Qian et al. [14]	MRI	–	ProCDet	TPR 91.82%
Wang et al. [15]	Clinical	–	LGBM	AUC 0.93
Xiao et al. [17]	Clin (EMR)	551	LR, DT, RF, SVM	77.1%
Barlow et al. [18]	PLCO Clin	–	PoPC (QDA)	91.6%
Savla et al. [19]	WSI patches	86,250	DenseNet + Clf.	94.9%
El Kholy & Elgohary [20]	Clinical	1,400	Random Forest	91%
Kurogi et al. [21]	WSI H&E	295,340	DiagSet (DL)	94.6%
Proposed Model	Clinical	(100)	RF, GB, XGB, LGBM	96%

V. CONCLUSION

Prostate cancer must be accurately and promptly predicted in order to be effectively managed and to enhance patient outcomes. In this work, we suggested an ensemble-based approach to improve prediction performance by utilizing soft voting and stacking. While the ensemble model performed better with 96% accuracy, balancing precision and recall, individual classifiers including XGBoost, Random Forest, LightGBM, SVM, and Logistic Regression produced good results with accuracies ranging from 82% to 92%. By emphasizing feature significance and model decision processes, the use of Explainable AI (XAI) approaches like SHAP and LIME further enhanced transparency and bolstered clinician trust. Furthermore, the suggested approach preserved computing efficiency, which qualifies it for practical healthcare applications where speed and precision are crucial. The remarkable outcomes demonstrate the potential of ensemble learning in cancer diagnosis, providing a potent instrument to support physicians in making decisions early on. To further improve predictive accuracy, future studies will concentrate on adding multi-omics data and patient-specific biomarkers, investigating deep learning ensembles and hybrid feature selection, and expanding the framework for more clinical relevance using enhanced XAI analysis and real-time prediction pipelines.

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